Given a categorical distribution :

$$\boxed{P(X = \text{select } i \text{ class}) = \pi_i}$$

Each class is associated with an independent G

Prove that : $\underbrace{Y}_{\text{random variable}} = \operatorname{argmax}(\log \pi_i + \underbrace{G_i}_{\text{random variable}})$

$$P(Y = i) = \pi_i \rightarrow \text{"max-stable" property}$$

where $G_i = -\log(-\log U_i)$

where $U_i = \text{Uniform}(0, 1)$

Some property :

CDF : $F(g) = P(G \leq g)$

Cumulat

$$= P(-\log(-\log U) \leq g)$$

$$= P(-\log U \geq e^{-g})$$

$$= P\left(U \le e^{-(e^{-g})} \right)$$

$$= e^{-(e^{-g})} \quad - 1$$

(the probability of $U \le e^{-e^{-g}}$ is $e^{-e^{-g}}$)

PDF: $f(g) = \dfrac{d\,F(g)}{dg}$

$$= \left(e^{-e^{-g}}\right) \cdot \frac{d\,-e^{-g}}{dg}$$

$$= e^{-e^{-g}} \cdot \left(-e^{-g}\right) \cdot (-1)$$

$$= e^{-e^{-g}} \cdot e^{-g} = e^{-(g + e^{-g})}$$

Let's compute the new random variable Y's probabity

$$P(Y = i) = P\left(\boxed{\log \pi_i + G_i \ge \log \pi_j + G_j \text{ where } j \ne i} \right)$$

$$= P(X_i \ge X_j \text{ where } j \ne i)$$

Let $\underline{X_i} = \log \pi_i + \underline{G_i}$ Note: $X_1, X_2, X_3 \ldots$ are i.i.d.

random variables

First try to understand X_i's CDF and PDF

CDF: $P(X_i \le x) = P(\log \pi_i + G_i \le x)$

$$= P(G_i \le x - \log \pi_i)$$

from ①

$$= e^{-e^{(\log \pi_i - x)}} = e^{-\pi_i e^{-x}} \quad ②$$

PDF: $P(X_i = x) = \dfrac{d\,P(X_i \le x)}{dx}$

$$= \frac{d\, e^{-\pi_i e^{-x}}}{dx}$$

$$= e^{-\pi_i e^{-x}} \cdot \frac{d - \pi_i e^{-x}}{dx}$$

$$= e^{-\pi_i e^{-x}} \cdot (-\pi_i e^{-x})(-1)$$

$$= \pi_i e^{-(\pi_i e^{-x} + x)}$$

The goal is to compute:

$$P(Y=i) = P(X_i \geq X_j,\ i \neq j)$$

$$P(X_i \geq X_j,\ i \neq j)$$

select $X_i = x$ first

$$= \int_{-\infty}^{\infty} P(X_i \geq X_j,\ i \neq j \mid \underline{X_i = x})\, P(X_i = x)\, dx$$

plug it in

$$= \int_{-\infty}^{\infty} P(x \geq X_j,\ i \neq j)\, P(X_i = x)\, dx$$

X_j are all i.i.d., we can decouple them

$$= \int_{-\infty}^{\infty} \underbrace{\prod_{j \neq i} P(X_j \leq x)}_{\text{CDF from ②}}\, P(X_i = x)\, dx$$

$$= \int_{-\infty}^{\infty} \left(\prod_{j \neq i} e^{-\pi_j e^{-x}} \right) \pi_i \, e^{-(\pi_i e^{-x} + x)} \, dx$$

$$= \pi_i \int_{-\infty}^{\infty} e^{\sum_j (-\pi_j e^{-x})} \, e^{-x} \, dx$$

take this out

$$= \pi_i \int_{-\infty}^{\infty} e^{-e^{-x} \cdot \boxed{\sum_j \pi_j}} \cdot e^{-x} \, dx$$

$\sum_j \pi_i = 1$, assume normalized

$$= \pi_i \int_{-\infty}^{\infty} e^{-e^{-x}} \cdot \underbrace{e^{-x} \, dx}$$

Let $u = e^{-x} \Rightarrow du = e^{-x}(-dx)$

$$= \pi_i \int_{0}^{\infty} (-1) \, e^{-u} \, du$$

$$= -\pi_i \int_0^{\infty} e^{-u} \, du$$

$$= -\pi_i \left(e^{-\infty} - e^{0} \right)$$

$$= -\pi_i \left(0 - 1 \right) = \pi_i \quad \#$$

To summarize:

$$P(Y=i) = P(X_i \geq X_j, \; i \neq j)$$

$$= \pi_i$$

We can see that adding G_i pertubation to $G_i + \log \pi_i$ plus argmax doesn't change the distribution